From lexical search to semantic matching: a multi-model architecture for SNOMED CT concept retrieval

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DRAFT, September 4, 2026

Abstract

Objectives. Reference terminologies such as SNOMED CT are challenging to search, although fast and responsive terminology servers allow clinicians to learn effective search strategies and iteratively refine their queries over SNOMED CT's curated descriptions. Some implementations require a more natural search experience that depends less on the user's terminology-search skills. The conventional response is an additional, locally curated interface vocabulary, but such vocabularies are labour-intensive to maintain and their quality can degrade over time. We investigate a generalizable alternative: combining query interpretation, complementary lexical and semantic retrieval, rank fusion and optional re-ranking to facilitate direct search over SNOMED CT's existing lexical content.

Materials and Methods. We implemented one reproducible instance of this architecture over the SNOMED CT International edition. It combines deterministic, order-independent multi-prefix search with three learned components: BioLORD-2023-M embeddings for semantic retrieval, an optional BGE cross-encoder for re-ranking, and a local LLM (gemma 3) for query normalization and structured clinical entity extraction. Lexical and semantic rankings are combined with Reciprocal Rank Fusion, and results can be constrained with a hierarchy (descendant) filter. As an external evaluation we measured search-only linking accuracy on 542 disease mentions from the DisTEMIST corpus (Spanish, zero-shot), resolving gold codes through SNOMED CT historical associations and ablating query pre-processing and re-ranking.

Results. Illustrative tests showed complementary behaviour: the lexical and semantic paths resolved different query types, while LLM preprocessing supported cross-lingual and abbreviated input. On a search-only linking evaluation over 542 disease mentions from the DisTEMIST corpus (Spanish, zero-shot), the best and computationally cheapest configuration — semantic search with re-ranking and *no* LLM query preprocessing — placed the exact concept first for 60% of mentions (accuracy@1 = 0.60) and within the top-10 candidates a picker would show for

80% (recall@10 = 0.80), in the range of the supervised systems from the original shared task. LLM query pre-processing helped dirty, lay input but slightly hurt clean terminology-like queries, and a near-miss analysis showed that many remaining cases retrieved a parent or child of the intended concept. Results are specific to the evaluated configuration, not an architecture-independent estimate.

Discussion. The findings support the technical feasibility of a reference architecture that computes additional access routes over SNOMED CT's curated descriptions. This shifts rather than eliminates maintenance, from local phrase authoring toward governance and validation of retrieval services.

Conclusion. Hybrid concept retrieval offers an implementation pattern that may reduce additional local lexical curation where direct search is appropriate, while curated interface vocabularies remain useful in selected guided workflows.

Background and significance

The vocabulary challenge

Reference terminologies such as SNOMED CT are challenging to search. They are engineered for concept permanence, formal definition and computability (Cimino 1998), whereas clinicians use compact, variable and often idiosyncratic expressions. This vocabulary challenge is inherent to human communication: when people name the same thing, they choose the same word less than 20% of the time (Furnas et al. 1987).

Terminology developers and implementers have addressed this challenge through different strategies. Efficient lexical algorithms improve access to the descriptions already curated within the terminology. When that is insufficient, interface terminologies add locally curated expressions intended to anticipate how users may search. More recently, language models have made it possible to compute semantic similarity at query time instead. These approaches place the work of resolving the vocabulary mismatch, together with its corresponding maintenance burden, in different places: within the reference terminology, with the local implementer or in the retrieval system. Our hypothesis is that dynamic semantic matching can replace much of the additional local lexical curation while continuing to rely on SNOMED CT's maintained descriptions.

Curated descriptions and algorithmic lexical search

SNOMED CT already includes substantial curated lexical content: its concepts are accompanied by clinician-authored descriptions, including preferred terms and synonyms (approximately 1.6M active descriptions for approximately 368k concepts in the 2024 International edition) (SNOMED International 2024). This content reflects two important development traditions. Clinical Terms Version 3 (CTV3, formerly the Read Codes) was explicitly user-led and stressed the inclusion of the natural terms used by clinicians (O'Neil et al. 1995). In parallel, SNOMED RT was developed through a collaborative process involving the College of American Pathologists and Kaiser Permanente's Convergent Medical Terminology (CMT) team (Levy et al. 1998). The merger of CTV3 and SNOMED RT brought these traditions together in SNOMED CT. This is also the genealogy of the clinician-facing vocabulary embedded in SNOMED CT: a centrally curated interface resource distributed and maintained with the reference terminology, rather than separately by each implementation.

In routine use, modern terminology servers combine this lexical content with fast, responsive search. Part-word retrieval has a long lineage in UK clinical terminology systems. In a randomized crossover trial, general practitioners used the NHS Clinical Terminology Browser to search Read Codes and CTV3 by entering one or more words or part-words (Brown et al. 2003). Later tooling for Read-coded data formalized searches using multiple word-stubs occurring anywhere in a description (Olier et al. 2016). The *multi-prefix, any-order* technique used here follows this interaction pattern: users enter only the first characters of one or more words, and each fragment is matched against the beginning of a word in the description, regardless of word order. Thus, the fragments *myo inf* can retrieve *myocardial infarction*, and reversing them does not change the match. This tolerates incomplete words and natural variation in word order while preserving precision by requiring every entered prefix. Because inflected and derived forms often share their initial characters, it also provides lightweight morphological normalization across variations such as number and, in languages that mark it, gender. This benefit is language-dependent and does not amount to full lemmatization. Clinicians can start with only a few characters and refine them in response to the results until they reach the concept they intend. This simple interaction makes the user an active participant in resolving the vocabulary mismatch, without requiring every possible expression to be anticipated in advance.

Additional local lexical curation

Some implementations, however, need to reduce that dependence on individual search skills. Adoption constraints, limited training opportunities or the demands of a particular workflow may call for more natural queries and more guided concept selection. The conventional response is an additional *interface terminology*: a "systematic collection of health care-related phrases that support clinicians' entry of patient-related information" (Rosenbloom et al. 2006). Added to the descriptions distributed with SNOMED CT, these local vocabularies map further colloquial expressions, abbreviations and preferred terms to concepts in the reference terminology. Kaiser Permanente's CMT is a prominent example: it combined SNOMED CT with clinician-facing terms, mappings and context-specific subsets as an enterprise terminology resource (Dolin et al. 2004).

Published operational experience shows that this convenience is sustained by ongoing expert work. At the University of Nebraska Medical Center, a just-in-time process allowed clinicians to submit problems that were not yet represented in its 12,000-term lexicon. Evaluating an unlisted problem, finding an appropriate code, notifying the clinician and coding the record required an average of 30 minutes of expert coder time; unresolved terms were escalated to a vocabulary team (Warren et al. 1998). At Hospital Italiano de Buenos Aires, a terminology service developed through continuous user feedback recognized 140 ways to describe arterial hypertension and thousands of Spanish clinical expressions absent from a general-language dictionary. Its permissive policy improved user acceptance, but 67% of concepts in the problem-list subset required post-coordination; local mappings required additional manual work; and rules based on SNOMED CT hierarchies had to be reviewed with each new release (González Bernaldo de Quirós et al. 2018). At enterprise scale, early CMT tooling was designed to reconcile conflicting local enhancements developed by Kaiser Permanente and Mayo Clinic and to distribute updates compatible with those extensions (Campbell et al. 1996). These are successful systems, but their success depends on specialized teams, governance and tooling.

That work creates a substantial maintenance obligation for each implementation. Every additional interface term must be authored, mapped to a concept and kept aligned as the reference terminology evolves. More generally, terminology maintenance has been characterized as complex, resource-intensive and time-consuming, with increasing size requiring more extensive change management, validation, consistency checking, versioning and editing support (Bakhshi-Raiez et al. 2008). SNOMED International notes that a custom interface terminology is “challenging to maintain and keep up to date with new SNOMED versions, with risks of quality issues such as duplication and ambiguity,” and recommends, where feasible, using SNOMED CT’s own descriptions as the interface terminology (SNOMED International 2024). Without sustained maintenance, detailed local interface terminologies and the value sets and mappings that depend on them can fall out of alignment with evolving reference terminologies, introducing duplication, ambiguity and inconsistencies across versions (Bakhshi-Raiez et al. 2008; SNOMED International 2024; González Bernaldo de Quirós et al. 2018). The tension between flexible, clinician-friendly expression and structured, maintainable representation is long-recognized (Rosenbloom et al. 2011). However, the imbalance between unbounded expressive variety and finite local curation capacity makes an exhaustively curated additional interface vocabulary difficult to sustain.

Dynamic semantic matching

Recent language technologies offer another way to facilitate search without creating a separate local vocabulary. Biomedical text-representation models place synonyms and paraphrases close in a shared semantic space even without lexical overlap (Remy et al. 2024; Liu et al. 2021); this approach has powered semantic search over large clinical ontologies (Ngo et al. 2021) and unsupervised concept annotation against SNOMED CT (Abdulnazar et al. 2024). Crucially, semantic matching does not replace the lexical curation performed by SNOMED CT: embeddings are generated from its existing descriptions, so the terminology's curated lexical content anchors the semantic representation. Combined with LLMs that can normalize and translate clinician shorthand, these methods can dynamically bridge the gap between a natural query and those descriptions, potentially replacing much of the manual expansion with additional local synonyms.

Objective

We propose a generalizable reference architecture for retrieving SNOMED CT concepts from free-text clinical expressions, and illustrate it with a locally runnable, open-source implementation and a preliminary external evaluation on a Spanish gold-standard corpus. The architecture separates query interpretation, complementary lexical and semantic candidate generation, rank fusion, optional re-ranking, and terminology-aware constraints. Our aim is not to prescribe a particular technology stack or introduce a new state-of-the-art entity linker. Instead, we examine *hybrid concept retrieval* as an alternative implementation pattern between direct lexical search and an additional hand-maintained interface vocabulary. Publishing the reference implementation makes its design choices inspectable and allows others to adapt and evaluate the pattern in their own settings.

Materials and methods

Reference architecture

At an abstract level, the proposed architecture comprises five separable functions: (1) interpreting or normalizing the input query; (2) generating candidates through complementary lexical and semantic retrieval paths; (3) combining evidence from rankings whose scores may not be directly comparable; (4) optionally re-ranking a small candidate set with a more computationally intensive model; and (5) applying terminology-aware constraints and deterministic safeguards. The architecture does not depend on a particular LLM, embedding model, re-ranker, database or fusion algorithm. Implementers can substitute or omit components according to their languages, latency requirements, available infrastructure and governance constraints. Its central principle is to preserve the precision and responsiveness of lexical matching while adding a semantic route for expressions that do not share the terminology's wording. Figure 1 summarizes these functions and the flow between them.

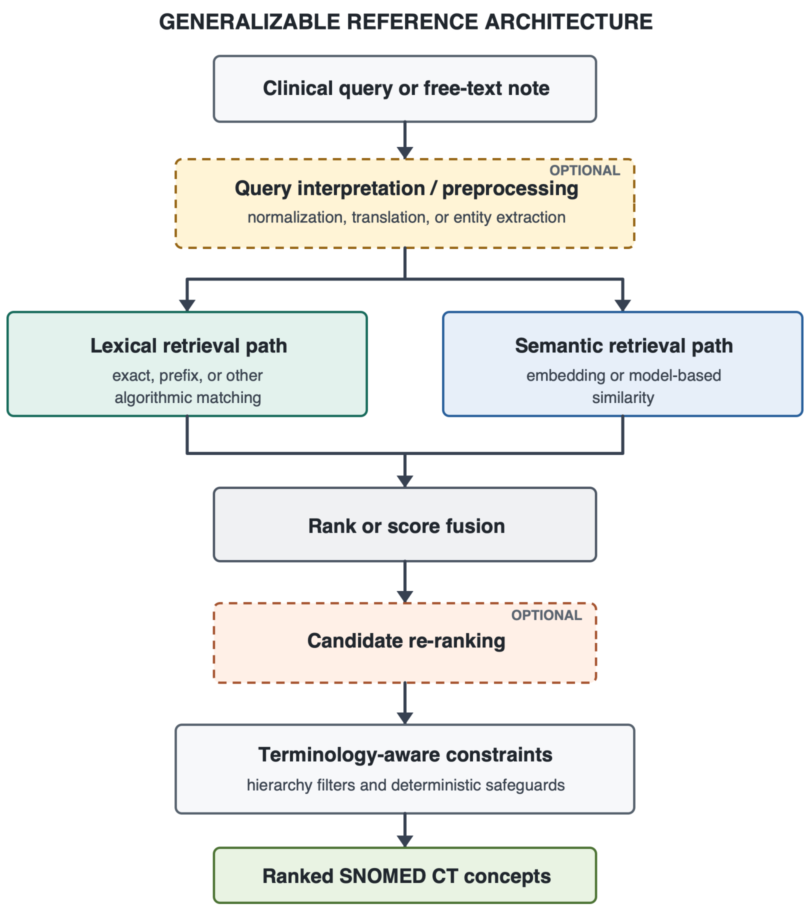

Figure 1: Generalizable reference architecture for hybrid SNOMED CT concept retrieval. Dashed borders indicate optional functions. Components within each function may be substituted according to local requirements; arrows show the flow of queries, candidates and rankings.

Reference implementation

The system evaluated here is one concrete instance of this model. A local LLM normalizes short queries or extracts clinical entities from longer text; deterministic multi-prefix matching supplies the lexical candidates; a biomedical bi-encoder supplies the semantic candidates; Reciprocal Rank Fusion combines both rankings; an optional cross-encoder re-ranks the fused candidates; and hierarchy filters constrain results to an appropriate SNOMED CT domain. In this implementation, retrieval, fusion and hierarchy filtering are served from a single PostgreSQL instance and executed in one query plan. The following sections describe these choices to make the example reproducible, rather than as requirements of the general architecture. The open-source implementation also provides an executable reference in which components can be inspected, replaced and re-evaluated against different languages, SNOMED CT editions, corpora and infrastructure.

Lexical channel

Active descriptions are indexed as PostgreSQL full-text vectors with a custom accent- and case-insensitive, non-stemming configuration and a GIN index. At query time, each fragment entered by the user becomes a prefix clause and the clauses are joined by AND; every fragment must therefore match the initial characters of a word in the description, in any order. For example, *myo inf* is represented as `myo:* & inf:*` and matches *Myocardial infarction* as readily as *inf myo*. Shared prefixes also absorb some language-dependent inflectional and derivational variation without explicit stemming or lemmatization. This supports incremental search with minimal typing while avoiding the broad result sets produced by matching only some of the entered prefixes. Ranking uses `ts_rank` with length and unique-word normalization so that concise canonical descriptions outrank verbose ones (PostgreSQL Global Development Group 2026).

Semantic channel

Each description is embedded with BioLORD-2023-M, a biomedical, multilingual sentence bi-encoder (Remy et al. 2024) whose lineage traces to self-aligned biomedical representations (Liu et al. 2021), and stored in a `pgvector` column with an HNSW index for approximate nearest-neighbour search by cosine similarity.

Fusion and re-ranking

The two rankings are combined with Reciprocal Rank Fusion (RRF) (Cormack et al. 2009), which requires no score calibration between channels. An optional cross-encoder re-ranker (BGE-reranker-v2-m3, built on a multilingual encoder (Chen et al. 2024)) refines the fused top-k; its score is min-max blended with the RRF score rather than allowed to override it. A deterministic rule pins exact-match descriptions to the top.

Hierarchy filter

A precomputed transitive closure of the IS-A graph allows any search to be constrained to the descendants-or-self of a concept (e.g. only *Clinical findings*), turning "is concept C under X?" into a single indexed array-containment lookup.

LLM roles

A local LLM (gemma 3 (Gemma Team, Google DeepMind 2025), served through an OpenAI-compatible endpoint) plays two roles. (1) *Query normalization*: it translates a short clinician query from any language to one canonical English clinical term and expands abbreviations, so that both retrieval channels operate on clean English; the embedding model's cross-lingual nature also allows the system to degrade gracefully without the LLM. (2) *Entity extraction*: given a free-text note, it returns structured entities as JSON, including the verbatim span, semantic type, assertion/temporal context (present/absent/past/unknown), laterality, severity and an English `clinicalTerm`. Each extracted entity is then mapped independently through the hybrid engine, using its normalized `clinicalTerm` as the query and its semantic type as the hierarchy filter (falling back to an unfiltered search when the type filter would exclude everything).

Data and implementation

We used the RF2 Snapshot view of the 1 June 2026 SNOMED CT International Edition production release. We loaded the active English-language descriptions of active concepts: 1,024,825 descriptions (381,856 fully specified names and 642,969 synonyms) for 381,856 concepts; of these, 1,020,940 unique normalized description strings were embedded. A transitive closure of the IS-A relationships was precomputed over all 381,856 active concepts to support the hierarchy (descendant) filter. The stack runs locally on a single workstation (Apple Silicon), with the database in a container and all model inference in-process; no clinical data leaves the machine. The full pipeline, indexes and demonstration interface are reproducible from a public repository [TODO: URL].

External evaluation

Choice of corpus. Openly licensed gold standards that link *free clinical text* to SNOMED CT are scarce. Most English clinical-note resources that normalize to SNOMED CT or the UMLS require a data-use agreement, and general clinical NLP corpora do not target SNOMED CT. The Spanish corpora produced by the Barcelona Supercomputing Center for the BioASQ/BioCreative shared tasks are a notable exception: they are released under CC BY 4.0 and every mention is normalized to SNOMED CT. We evaluated on **DisTEMIST** (BioASQ 2022)

— 1,000 Spanish clinical case reports with disease mentions linked to SNOMED CT (Miranda-Escalada et al. 2022) — because diseases map cleanly onto the *Clinical finding* hierarchy; the companion **SympTEMIST** (symptoms, signs and findings) (Lima-López et al. 2023) shares the same format. Two caveats follow from this choice and we treat them explicitly rather than hide them. First, the corpus is **Spanish**; for a multilingual system this is less a limitation than a cross-lingual stress test, since the query and the terminology descriptions are in different languages. Second, the texts are drawn from **published case reports**, whose language is more edited than bedside notes; the evaluation therefore probes concept normalization more than the full messiness of real electronic health record text. We use this corpus because it is among the few open, SNOMED-CT-linked options, and read the results with these limits in mind.

Task and configurations. To measure the *search* capability independently of our entity extractor, we ran a **search-only** evaluation: each gold mention's verbatim span is submitted as a query and we record the rank of the gold SNOMED CT code — this is the corpus's own entity-linking (normalization) subtask. We swept the two learned, optional components of the search pipeline as a 2×2 design — LLM query pre-processing (gemma normalization/translation) on/off × cross-encoder re-ranking on/off — and report three of the four cells. The lexical and semantic channels and their RRF fusion were always active; the hierarchy filter was fixed to *Clinical finding* (the corpus is diseases); and we retrieved $k = 10$ candidates.

Edition-drift resolution. The gold was annotated against a circa-2022 edition; by our loaded release (International 2026-06-01) some gold concepts had been inactivated. Because a fixed gold standard drifts as the terminology evolves — itself an instance of the maintenance problem this paper is about — we resolved every gold code through SNOMED CT's **historical-association** reference set (in priority order SAME AS → REPLACED BY → POSSIBLY REPLACED BY → POSSIBLY EQUIVALENT TO → ALTERNATIVE) to its current active concept before scoring.

Scope. The system targets *disorder* and *finding* concepts. Where the annotators coded a mention to a *substance* or *morphologic-abnormality* concept for the same clinical entity (for example, *renal stone* as a substance rather than as a disorder), a system that returns the disorder can never match by identifier. We therefore scored only gold whose resolved concept is a disorder or finding, so that representation choices outside the system's target do not create artificial misses.

Metrics. We chose metrics that mirror how a clinician uses a concept picker. In a typical search interface the user types a term and selects from a short ranked list, so what matters is whether the intended concept appears *near the top*. We therefore report, under a **strict** rule (our concept identifier equals the resolved gold): **accuracy@1** — the top candidate is exactly the gold concept; **recall@5** and **recall@10** — the gold concept is among the first 5 or 10 candidates, i.e. the options a user would see in a drop-down list without, or with a small scroll; and **mean reciprocal rank (MRR)**, which rewards a higher position of the gold concept. We deliberately foreground recall@5 and recall@10 because they answer the practical question for a picker interface — "would the right concept be on the screen for the user to choose?" — rather than requiring the system to be right on its single best guess. Separately, because a *parent* or *child* of the intended concept can be an acceptable or navigable alternative, we report a **near-miss** rate: the additional fraction of mentions (beyond the exact matches) whose top-k list contains an ancestor or descendant of the gold. The near-miss rate is thus an increment on strict recall, not a replacement for it — the two together sum to a "same-lineage" recall. It is an upper bound on "clinically acceptable" rather than a hard accuracy, since a hierarchical neighbour is not always an acceptable substitute; confirming that would require clinician adjudication.

Results

Feasibility of the reference architecture

The open-source reference implementation instantiated all five functions of the proposed architecture in an end-to-end pipeline. For short expressions, it generated lexical and semantic candidates and returned a fused ranking. For longer notes, it first extracted and normalized clinical entities and then retrieved candidates independently for each entity. Optional re-ranking and hierarchy constraints operated on the same candidate flow, allowing the architecture to support both interactive concept search and note-level entity linking. This establishes the technical feasibility of the architectural pattern; the observations and metrics below characterize this particular implementation.

Illustrative component behaviour

The examples exposed complementary roles for the components rather than a consistent advantage for any single model. The semantic path recovered lay phrasing with no shared tokens ("water on the knee" → *Effusion of knee joint*), whereas deterministic multi-prefix matching retrieved a clinician abbreviation ("inf myo" → *Myocardial infarction*) that was represented poorly by the embedding model. Query interpretation and cross-lingual retrieval supported inputs such as "radiografía de tórax" → *Plain X-ray of chest* and "EPOC" → *Chronic obstructive pulmonary disease.* For longer notes, the LLM extracted entities and represented assertion separately from concept identity, mapping "no fever" to *Fever* with an absent flag and "history of stroke" to *Cerebrovascular accident* with a past flag. In a Spanish note, entities were extracted verbatim, normalized to English `clinicalTerms`, and mapped to the expected SNOMED CT concepts (e.g. *diabetes mellitus tipo 2* → *Diabetes mellitus type 2*; *disnea de esfuerzo* → *Dyspnea on exertion*).

External evaluation on DisTEMIST

We evaluated the first 100 DisTEMIST training documents, which contain 588 unique disease mentions; 20 gold codes were recovered through historical associations and 46 mentions were scoped out as non-disorder/finding, leaving **542** mentions. Table 1 reports the three search configurations.

Table 1: Search-only linking on 542 DisTEMIST disease mentions (Spanish, zero-shot). "Pre-proc." = LLM query normalization; "Re-rank" = cross-encoder. Metrics use the strict rule (exact concept identifier); recall@5 and recall@10 are the picker-list framing (is the gold concept on the screen the user would see?). *near-miss@10* is the additional fraction whose top-10 held an ancestor or descendant of the gold. Best strict value per column in **bold**.

Pre-proc.	Re-rank	acc@1	recall@5	recall@10	MRR	near-miss@10
off	**on**	**0.60**	**0.76**	**0.80**	**0.67**	+0.09
on	off	0.54	0.70	0.75	0.60	+0.12
on	on	0.54	0.69	0.75	0.61	+0.12

Read the numbers as a concept picker: recall@5 and recall@10 are the chance that the intended concept is on the short list a clinician would scan and select from, while accuracy@1 is the chance the system's first candidate is already correct. Three findings stand out. First, the **cheapest configuration is also the best**: with query pre-processing *off* and re-ranking *on*, the exact concept is the top candidate for 60% of mentions (accuracy@1 = 0.60) and appears within the top-10 picker list for 80% (recall@10 = 0.80); for a further 9% of mentions a parent or child of the intended concept is in that list (near-miss@10 = +0.09). Because this configuration makes no per-query LLM call, it is simultaneously the most accurate and the least expensive to run. Second, **LLM query pre-processing hurts on this task**: enabling it lowers accuracy@1 from 0.60 to 0.54. The gold mentions are already clean, terminology-like disease terms, and the multilingual embedding maps them cross-lingually without help; the LLM's normalization to English instead shifts the semantic neighbourhood (for example, *infertilidad* → *infertility* pulls the specific "X infertility" disorders above the exact generic *Infertile*). The LLM's value is for the dirty, lay or abbreviated input of interactive use, not for well-formed terminology strings. Third, **re-ranking helps** when it is not preceded by aggressive normalization, improving both accuracy@1 and MRR; the two components are genuinely optional and the optimal configuration is task-dependent.

These are **zero-shot** results — no training or fine-tuning on the corpus — yet accuracy and recall fall in the range reported by the *supervised* systems of the original shared task (Miranda-Escalada et al. 2022). The near-miss rate shows that a meaningful share of the strict "errors" are not wrong answers but hierarchically adjacent ones: at rank 1, beyond the 0.60 exact matches, a further

0.15 of mentions have a parent or child concept as the top candidate, and across the top-10 the near-miss increment is 0.09–0.12. Whether such a neighbour is acceptable is a clinical judgement, so we treat this as an upper bound rather than as accuracy. A further residual class, credited by neither the exact nor the near-miss count, is the annotators' choice of a *substance* or *morphologic-abnormality* concept where the system returns the corresponding *disorder* — a genuine representation difference in the gold itself, not a retrieval failure.

Running the full extraction-then-mapping pipeline over the same notes (rather than the gold spans) lowers these figures, which isolates the contribution of the extraction stage: the LLM occasionally mis-types or omits an entity, and long notes must be chunked to prevent the small model from degenerating and dropping their tail. In this implementation the search is the stronger component; extraction is where accuracy is currently lost.

Discussion

Principal findings. This study contributes a generalizable reference architecture and an executable, open-source implementation. The implementation demonstrates the technical feasibility of combining query interpretation, complementary lexical and semantic retrieval, rank fusion, optional re-ranking and terminology-aware constraints in a single concept retrieval flow. Illustrative examples showed that these functions address different forms of vocabulary mismatch, while the preliminary external evaluation established a reproducible baseline for one concrete configuration. These findings support the coherence and feasibility of the architectural pattern. They do not establish that the selected components are optimal, that the observed performance transfers to other implementations, or that the system is ready for clinical use.

Implications for terminology infrastructure. Hybrid concept retrieval provides an intermediate strategy between expecting clinicians to resolve every mismatch through lexical query refinement and attempting to anticipate every likely expression in an additional interface vocabulary. A responsive lexical path preserves familiar, precise interaction with SNOMED CT descriptions, while a semantic path computes further candidates when the query uses different wording. Both paths remain anchored in lexical content curated and distributed with SNOMED CT. This may reduce local term-by-term authoring and mapping, but it does not make maintenance disappear. Instead, it shifts work toward governing models, indexes, constraints, terminology releases and regression tests. Some of this work can be automated and shared, whereas exhaustive local phrase curation grows with the variety of clinician language (Furnas et al. 1987). Additional curated interfaces remain appropriate when highly guided workflows, local policy or limited opportunities for user training make them valuable. Using SNOMED CT's own descriptions directly where feasible also follows SNOMED International guidance (SNOMED International 2024).

Architecture, implementation and validation. The five architectural functions define separable responsibilities rather than fixed products. Implementations may use different language models, embedding models, lexical algorithms, fusion methods, databases and constraints, and their results may therefore differ substantially. Performance claims should attach to a specified, versioned implementation, while support for the reference architecture should accumulate across implementations and settings. The open-source system and evaluation materials provide a starting point for a reusable evaluation harness: canonical queries, external corpora, ranking metrics, latency measures and failure cases can be rerun when a component changes. Automated regression testing can make iterative improvement less costly, but it cannot replace validation against the intended users and clinical use case. The required depth of that validation should reflect the consequences of an incorrect match and whether the output is advisory, user-confirmed or recorded without review.

Relation to prior work. Semantic search over clinical ontologies and SapBERT/BioLORD-style bi-encoders established that embeddings can recover concepts under vocabulary mismatch (Ngo et al. 2021; Liu et al. 2021; Remy et al. 2024; Abdulnazar et al. 2024), while recent scoping work catalogues uses of LLMs with SNOMED CT (VERIFY authors 2024). The distinctive contribution here is architectural rather than a new retrieval model. The design assigns bounded roles to an LLM for query interpretation and entity extraction, preserves a deterministic lexical path, adds semantic candidate generation, combines rankings without requiring score equivalence, permits more expensive re-ranking only over a small candidate set, and applies constraints derived from the terminology. It connects these technical choices to the operational question of whether access to SNOMED CT can be made more forgiving without creating another large vocabulary to maintain.

Limitations. The external evaluation covers one corpus, one language, one concept type (diseases) and the search-only linking task on gold mention spans; end-to-end extraction, other concept types (procedures, findings), other languages, and real electronic-health-record notes rather than published case reports remain to be tested. The corpus being Spanish and drawn from case reports means the input is cleaner than bedside documentation. The strict/hierarchy-aware gap indicates that the exact-identifier metric understates semantically acceptable retrieval, but neither metric was confirmed by blinded clinical adjudication. We ablated the two optional components (query pre-processing and re-ranking) but did not substitute the embedding model, lexical algorithm or fusion method, so the contribution of those functions is not estimated. We also did not compare clinician search performance, measure longitudinal maintenance effort, or evaluate safety in a production workflow. LLM entity typing remains imperfect, extraction latency is LLM-bound, and mapping to a single best concept omits post-coordination. Open source makes the design choices transparent and facilitates independent evaluation, but does not remove the need to validate each implementation against its intended users, terminology edition, language and use case. Next steps include larger evaluations across findings, diseases and procedures; component substitution and ablation experiments; hierarchy-aware and clinician-adjudicated metrics; usability studies; and longitudinal measurement of maintenance work.

Conclusion

SNOMED CT's curated lexical content can serve as the foundation for an interface that is partly *computed* at query time. The proposed reference architecture offers a testable, implementation-independent path for combining lexical precision with semantic flexibility. Its open-source implementation demonstrates feasibility and enables adaptation and re-evaluation, but each resulting system requires validation in its intended setting. This approach may reduce additional local lexical curation, while hand-built interface terminologies retain a role in guided, low-friction clinical workflows.

Acknowledgements

OpenAI Codex was used during manuscript preparation to assist with language editing, structural revision and drafting. All AI-assisted content was critically reviewed, verified and revised by the authors, who take full responsibility for the final manuscript.

Funding

[TBD]

Competing interests

[TBD]

Data and code availability

The reference implementation, configuration and evaluation materials are available as open-source software [TODO: repository URL]. They are provided to make the architectural choices inspectable and to support adaptation and re-evaluation in other environments. DisTEMIST (Miranda-Escalada et al. 2022) and SympTEMIST (Lima-López et al. 2023) are available under CC BY 4.0 from their authors and are not redistributed here.

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